

Hongzong LI

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RESEARCH INTERESTS

Optimization, Neural Networks, Large Language Model, Computational Intelligence, Matrix Factorization, Clustering

EMPLOYMENT

Hong Kong University of Science and Technology, Clear Water Bay, Hong Kong

Jul. 2025 - Present

Postdoctoral Fellow, Hong Kong Generative AI Research and Development Center

- Advisor: [Prof. Dapeng WU](#) (Chair Professor, IEEE Fellow)
- Leveraging Hopfield networks to improve associative thinking capabilities in large language models (LLM).

City University of Hong Kong, Kowloon, Hong Kong

Feb. 2025 - Jun. 2025

Postdoctoral Fellow, Department of Computer Science

- Advisor: [Prof. Dapeng WU](#) (Chair Professor, IEEE Fellow)

EDUCATION

City University of Hong Kong, Kowloon, Hong Kong

Jan. 2021 - Dec. 2024

Ph.D. in Computer Science

- Advisor: [Prof. Jun WANG](#) (Member of Academia Europaea, IEEE Life Fellow, HKAE, FIAPR)
- Developing optimization algorithms for capacitated clustering, quadratic unconstrained binary optimization, binary matrix factorization, etc.
- Developing clustering algorithms and applying clustering algorithms to finance and biology

Northeastern University, Shenyang, China

Sept. 2016 - July 2020

B.E. in Automation, School of Information Science and Engineering

- Overall GPA: 3.9891/5 - ranking within the top 5%.
- Advisor: [Prof. Dong XIAO](#)
- Focusing on the applications of machine learning (Published: 8 papers, 5 student first-authored papers, and 8 filed patents and software copyrights)

HONORS & AWARDS

Doctor of Philosophy (Ph.D.)

- Outstanding Academic Performance Award, 16 Aug. 2024
- Outstanding Academic Performance Award, 16 Aug. 2023
- Postgraduate Studentship (18,270 HKD per month)
- Institutional Research Tuition Scholarship (3,508 HKD per month)
- Institutional Research Tuition Grant (3,508 HKD per month)

Bachelor of Engineering (B.E.)

- Outstanding Graduates of Liaoning Province in 2020
- The Special Prize of excellent students of Baosteel in 2019 (20,000 CNY)
- The Fourth most influential graduate of the College in 2020
- The First Prize of "TI" Cup Electronic Design Competition for undergraduate students in Liaoning Province
- The Special Prize of the 7th China TRIZ Cup Undergraduate Innovation Method Competition
- The First Prize of the fourth Liaoning "TRIZ Cup" college students innovation method competition
- Northeastern University (2019) National Innovation Training Program for College Students - National Excellent
- Northeastern University (2019) National Innovation Training Program for College Students - my favorite innovation program for College Students
- Northeastern University (2018) National Innovation Training Program for College Students - Provincial Qualification
- The Second Prize of the 17th Undergraduate Electronic Design Competition of "Jianlong iron and steel" of Northeastern University in 2018
- The Third Prize of the fourth Internet plus China Northeastern University Student Competition
- First-class scholarship for outstanding students at Northeastern University (2018-2019, and 2019-2020)
- Third-class scholarship for outstanding students at Northeastern University (2016-2017, and 2017-2018)

- [1] **H. Li** and J. Wang, "From Soft Clustering to Hard Clustering: A Collaborative Annealing Fuzzy c-means Algorithm," *IEEE Transactions on Fuzzy Systems*, vol. 32, pp. 1181-1194, 2024. (IF: 12.253)
- [2] **H. Li** and J. Wang, "Capacitated Clustering via Majorization Minimization and Collaborative Neurodynamic Optimization," *IEEE Transactions on Neural Networks and Learning Systems*, vol. 35, pp. 6679-6692, 2024. (IF: 14.255)
- [3] **H. Li** and J. Wang, "Machine-Cell and Part-Family Formation via Neurodynamics-Driven Constrained Binary Matrix Factorization," *IEEE Transactions on Systems, Man, and Cybernetics: Systems*, 2025. (IF: 8.7)
- [4] **H. Li** and J. Wang, "A Collaborative Neurodynamic Algorithm for Quadratic Unconstrained Binary Optimization," *IEEE Transactions on Emerging Topics in Computational Intelligence*, vol. 9, pp. 228 - 239, 2024. (IF: 5.3)
- [5] **H. Li**, J. Wang, N. Zhang, and W. Zhang, "Binary Matrix Factorization via Collaborative Neurodynamic Optimization," *Neural Networks*, vol. 176, pp. 106348, 2024. (IF: 7.8)
- [6] **H. Li** and J. Wang, "CAPKM++ 2.0: An Upgraded Version of the Collaborative Annealing Power K-means++ Clustering Algorithm," *Knowledge-Based Systems*, vol. 262, p. 110241, 2023. (IF: 8.139)
- [7] **H. Li** and J. Wang, "Collaborative Annealing Power K-means++ Clustering," *Knowledge-Based Systems*, vol. 255, p. 109593, 2022. (IF: 8.139)
- [8] **H. Li** and J. Wang, "Co-clustering for Binary Data via Neurodynamics-driven Binary Matrix Factorization," in *13-th International Conference on Intelligent Control and Information Processing (ICICIP)*. IEEE, 2025, pp. 90-97.
- [9] **H. Li**, J. Wang, and J. Wang, "Solving the Travelling Salesman Problem Based on Collaborative Neurodynamic Optimization with Discrete Hopfield Networks," in *11-th International Conference on Information Science and Technology (ICIST)*. IEEE, 2021, pp. 456-465.
- [10] **H. Li** and J. Wang, "A Collaborative Neurodynamic Optimization Algorithm Based on Boltzmann Machines for Solving the Traveling Salesman Problem," in *11-th International Conference on Intelligent Control and Information Processing (ICICIP)*. IEEE, 2021, pp. 325-333.
- [11] **H. Li** and J. Wang, "Collaborative Neurodynamic Algorithms for Solving Sudoku Puzzles," in *12-th International Conference on Information Science and Technology (ICIST)*. IEEE, 2022, pp. 8-17.
- [12] **H. Li** and J. Wang, "A Collaborative Neurodynamic Optimization Algorithm Based on Boltzmann Machines and 2-Opt Heuristic for Solving the Traveling Salesman Problem," in *17-th International Conference on Advanced Computational Intelligence (ICACI)*. IEEE, 2025, PP. 1-8.
- [13] X. Ye*, **H. Li*** and J. Wang, "HVAC System Fault Diagnosis via Feature Selection and Classification," in *13-th International Conference on Information Science and Technology (ICIST)*. IEEE, 2023, pp 432-440.
- [14] D. Xiao, **H. Li**, and X. Sun, "Coal Classification Method Based on Improved Local Receptive Field-Based Extreme Learning Machine Algorithm and Visible-Infrared Spectroscopy," *ACS Omega*, vol. 5, no. 40, pp.25 772-25 783, 2020. (First student author)
- [15] D. Xiao, **H. Li**, C. Liu, and Q. He, "Large-Truck Safety Warning System Based on Lightweight SSD Model," *Computational Intelligence and Neuroscience*, vol. 2019, 2019. (First student author)
- [16] D. Xiao, **H. Li**, B. T. Le, S. Zhang, J. Wang, D. He, and X. Fu, "Research on a Method of Gross Error Elimination for Slope Monitoring Data Based on Machine Learning," *IEEE Access*, vol. 7, pp. 164 682-164 695, 2019. (First student author)
- [17] D. Xiao, **H. Li**, Z. Ji, E. Xu, B. Luo, and J. Chen, "An Anti-Collision Early Warning System for Mine Trucks Based on RBF Network and Wi-Fi," *Journal of Physics: Conference Series*, vol. 1631, p. 012157, 2020. (First student author)
- [18] D. Xiao, **H. Li**, and G. Jiang, "Spectral and BP Neural Network Research on Classification of Iron Ore," in

ACADEMIC SERVICE

Publication chair and committee member: The 17th International Conference on Advanced Computational Intelligence (ICACI2025), Bath, UK.

Session chair and committee member: The 13th International Conference on Information Science and Technology (ICIST2023), Cairo, Egypt.

Reviewer of the following journals and conferences: • IEEE Transactions on Pattern Analysis and Machine Intelligence, • IEEE Transactions on Neural Networks and Learning Systems, • IEEE Transactions on Knowledge and Data Engineering, • IEEE Transactions on Industrial Electronics, • IEEE Transactions on Consumer Electronics, • IEEE Transactions on Computational Social Systems, • Neural Networks, • Mathematical Biosciences and Engineering, • Signal Processing, • Journal of Low Frequency Noise, Vibration & Active Control, • Science China Technological Sciences, • Cloud Computing and Data Science, • ICIST2022, • ICIST2024, • ICICIP2024, • ISSN2024, • ICACI2025, • ICICIP2025, • ICCA2024, • IJCNN2025, • NeurIPS2024, • AAAI2025, • ICLR2025, • AISTATS2025, • ICML2025, • AAAI2026, • AISTATS2026.

MENTORING EXPERIENCE

Ran Zhang <i>Master at City University of Hong Kong</i> ➤ Topic: Portfolio and Index Tracking	Jun. 2021 - Dec. 2021
Xuntan Ye <i>Undergraduate at City University of Hong Kong</i> ➤ Topic: Classification for HVAC System Fault Diagnosis	Jan. 2023 - Dec. 2023

TEACHING EXPERIENCE

Teacher (Tutorial) in CS5486 Intelligent Systems at CityU (DG), Dong Guan	Spring 2025
Teaching assistant in CS5487 Machine Learning: Principles and Practice at CityU, Hong Kong	Fall 2024
Teaching assistant in CS4386 AI Game Programming at CityU, Hong Kong	Spring 2024
Teaching assistant in CS5489 Machine Learning: Algorithms and Applications at CityU, Hong Kong	Fall 2023
Teacher (Tutorial) in CS5486 Intelligent Systems at CityU, Hong Kong	Fall 2022
Teaching assistant in CS4386 AI Game Programming at CityU, Hong Kong	Spring 2023
Teaching assistant in GE2313 Global IT Case Studies at CityU, Hong Kong	Spring 2022, Fall 2022
Teaching assistant in CS1302 Introduction to Computer Programming at CityU, Hong Kong	Spring 2021

TALKS & PRESENTATIONS

- Co-clustering for Binary Data via Neurodynamics-driven Binary Matrix Factorization
In ICICIP2025, Muscat, Oman, Feb. 2025
- A Collaborative Neurodynamic Optimization Algorithm Based on Boltzmann Machines and 2-Opt Heuristic for Solving the Traveling Salesman Problem
In ISSN2024, Weihai, Shandong, Jul. 2024
- HVAC System Fault Diagnosis via Feature Selection and Classification
In ICIST2023, Cairo, Egypt, Dec. 2023
- Collaborative Neurodynamic Algorithms for Solving Sudoku Puzzles
In ICIST2022, Online, Oct. 2022
- Solving the Travelling Salesman Problem Based on Collaborative Neurodynamic Optimization with Discrete Hopfield Networks
In ICIST2021, Online, May 2021
- A Collaborative Neurodynamic Optimization Algorithm Based on Boltzmann Machines for Solving the Traveling Salesman Problem
In ICICIP2021, Online, Dec. 2021

PATENTS

Patents of Invention:

- Anti-collision method and device for mine car, Dong Xiao, **Hongzong LI**, Qifei He, CN110395207A, 2019.11.01
- Mine truck anti-collision warning system and method based on radar and WIFI, Dong Xiao, **Hongzong LI**, Qifei He, CN110459074A, 2019.11.15
- Method and device for detecting iron content in iron ore, Dong Xiao, Guotai Jiang, **Hongzong Li**, Zeyuan Zhang, CN109060675A, 2018.12.21

Software Copyrights:

- Vehicle panoramic assisted driving system, Dong Xiao, **Hongzong LI**, Qifei He, 2019SR0716144
- WIFI-based anti-collision warning system for mining trucks, Dong Xiao, **Hongzong LI**, Qifei He, 2019SR1029773

Patents of Utility Model:

- Anti-collision device for mine car, Dong Xiao, **Hongzong LI**, Qifei He, CN210653003U, 2020.06.02
- Mine truck anti-collision warning system based on radar and WIFI, Dong Xiao, **Hongzong LI**, Qifei He, Xiaotong Zhang, Yuan Lin, Chenyi Liu, CN210574333U, 2020.05.19
- A radar and GPS based collision avoidance device for mine cars, Qifei He, **Hongzong LI**, Jian He, CN212675166U, 2021.03.09

TECHNICAL SKILLS

Programming languages: MATLAB, Python, C++
ML/AI: Pytorch, Numpy, Pandas, Matplotlib

Web Technologies: HTML
Miscellaneous: Git, Shell, Latex, Unity